Exp.no:8
28.9.24

Nmap Live Host Discovery ARP, ICMP and TCP/UDP Ping scans

Aim:

To discover live hosts on a network using Nmap host discovery scans - and to understand which protocol operate at which layer of the TCP/IP model, and when each one applies

ARP Host Discovery:

$ sudo nmap -PR -sn to 200.6.0/24

O/P:

Starting Nmap 7.80 (https://nmap.org)
Nmap scan report for 10.200.6.1
Host is up.
Nmap done: 256 IP addresses (1 host up) scanned in 10.51 seconds

ICMP Host discovery:

7.1) ICMP Echo (-PE):

$ sudo nmap -PE -sn 10.200.6.0/24

Nmap scan report for 10.200.6.1
Host is up (0.069s latency)
Nmap scan report for 10.200.6.50

Host is up (0.07s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
N-map done: 250 IP address
12 hosts up
scanned in 8.44 seconds

ICMP Timestamp (-PP)

$nmap -PP -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.069s latency)
Nmap scan report for
10.200.6.250
Host is up (0.0695 latency)
Nmap done: 250 IP addresses
(2 hosts up)
scanned in 18.44 seconds.

ICMP Address Mask (-PM)

$nmap -PM -sn 10.200.6.0/24
Nmap done: 256 IP addresses
(0 hosts up) scanned in 52.14 seconds

TCP & UDP Discovery:

(i). TCP SVN Ping:

$nmap -PS -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50

Host is up (0.0705 latency)

Nmap done: 256 IP addresses (2 hosts up) scanned in 33.88 seconds

(ii) TCP Ack Ping (-PA)

$ sudo nmap -PA -sn 10.200.6.0/24

Nmap scan report for 10.200.6.50

Host is up (0.0705 latency)

Nmap scan report for 10.200.6.250

Host is up (0.0695 latency)

Nmap done: 256 IP addresses (2 hosts up) scanned in 33.64 seconds

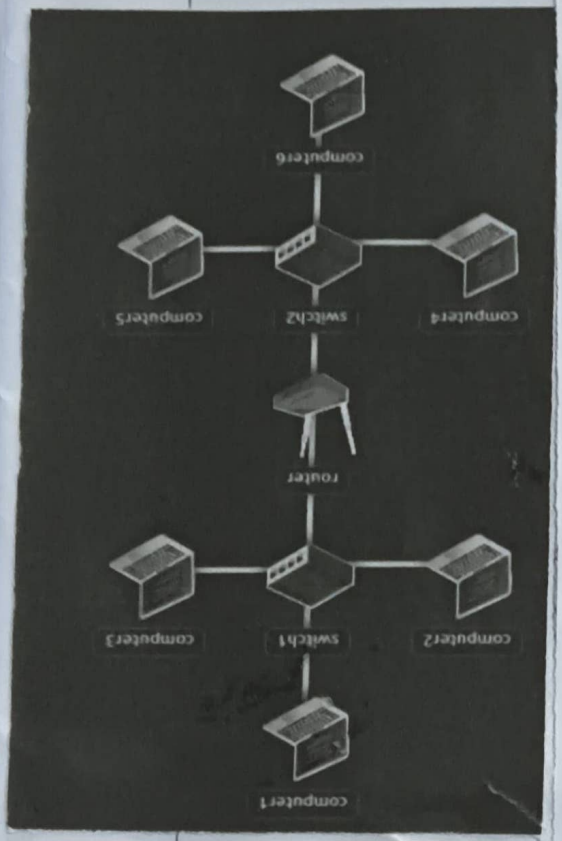
(iii). UDP Ping (-PU)

$ sudo nmap -PU -sn 10.200.6.0/24

Nmap scan report for 10.200.6.50

Host is up (0.0708 latency)

Nmap scan report for
10.200.6.250
Host is up (0.0683 latency)
Nmap done : 256 IP addresses
(2 hosts up) scanned in 35.97 seconds





Result:

Thus the Host discovery for ARP, ICMP, TCP, UDP was studied and the output got verified successfully.